
PROFILE

Computer Science undergraduate with hands-on experience in machine learning, NLP, LLM fine-tuning, RAG, automated scoring, and model evaluation. Experienced with Python, PyTorch, Hugging Face, FastAPI, and end-to-end AI system development. Seeking an internship in Machine Learning, NLP, or AI Engineering.

EDUCATION**VNUHCM – University of Information Technology, Bachelor of Science in Computer Science**

Sep 2023 – Jul 2026

- GPA: 3.40 / 4.0 | 8.27 / 10.0
- Relevant Coursework: Machine Learning, Advanced NLP, Data Mining, Artificial Intelligence, Deep Learning, Mobile - Web Programming
- Awarded a Certificate of Academic Excellence by the Rector of VNUHCM-UIT for outstanding academic performance in Term I 2023–2024, Term II 2024–2025, and Term I 2025–2026.

SKILLS

- **Programming:** Python, C++, Java, SQL, C#, Kotlin
- **Machine Learning / NLP:** PyTorch, TensorFlow, scikit-learn, Hugging Face, Transformers, LangChain, LangGraph
- **LLM / RAG:** LLM fine-tuning, Retrieval-Augmented Generation, prompt-based evaluation, model evaluation
- **Data / Visualization:** NumPy, Pandas, Matplotlib, Seaborn
- **Backend / Tools:** FastAPI, Gradio, Git, GitHub, MySQL, PostgreSQL, SQLite, Docker, MLFlow

LANGUAGES

English B2 – IELTS 6.5

EXPERIENCE**Computational Engineer Intern**Mar 2026 – May 2026 *CADFE Vietnam*

- Worked on engineering simulation workflows using PyAnsys for computational analysis and automation tasks.
- Integrated Python scripting with Ansys tools to improve simulation efficiency and reproducibility.
- Assisted in investigating simulation setups, automation pipelines, and technical issues in computational engineering applications.

PROJECTS**IELTS Writing Task 2 Automated Scoring System | GitHub**

2025 – 2026

- Developed an automated essay scoring system for IELTS Writing Task 2 using LLMs, NLP features, RAG, and transformer-based models.
- Fine-tuned models to predict four IELTS criteria: Task Response, Coherence and Cohesion, Lexical Resource, and Grammatical Range and Accuracy.
- Engineered linguistic and statistical features to improve scoring reliability, robustness, and interpretability.
- Built a RAG pipeline to retrieve similar essays and generate explainable feedback with strengths, limitations, and revision suggestions.
- Achieved approximately 0.70 mean QWK and 0.80 MAE on the test set.
- **Tech Stack:** Python, PyTorch, Hugging Face, Qwen, Mistral, SentenceTransformers, scikit-learn, FastAPI, Pandas, NumPy

Hate & Offensive Speech Detection using BERT | GitHub

2025 – 2026

- Built a BERT-based multi-class classifier for hate speech and offensive language detection on social text data.
- Prepared training, validation, and test data through tokenization, label mapping, and text normalization.
- Improved model performance through fine-tuning, error analysis, and iterative hyperparameter refinement.
- Analyzed misclassification patterns to identify difficult classes and improve robustness across labels.
- Achieved 95% accuracy and 0.88 macro-F1 on the test set.
- **Tech Stack:** Python, PyTorch, Hugging Face Transformers, BERT, scikit-learn, Pandas, NumPy

Vietnamese History RAG-SFT LLM with QLoRA | GitHub

2026 – Present

- Built a two-stage Vietnamese history LLM pipeline with Qwen2.5-Instruct: phase 1 for instruction tuning and phase 2 for RAG-SFT to teach evidence selection and grounded answering.
- Created and validated a RAG-SFT dataset with 1,000 chat-style samples from 520 Vietnamese history chunks, covering grounded QA, noisy-context, insufficient-context, and false-premise cases.
- Fine-tuned the model using 4-bit QLoRA, PEFT LoRA adapters, gradient checkpointing, weighted assistant-only loss, and early stopping based on generation metrics.
- Designed source-aware training so the model outputs selected chunk_id evidence before answering, improving grounded response behavior in noisy RAG contexts.
- Achieved strong validation generation results: **0.96 source F1, 0.96 exact source match, 1.00 format correctness, 1.00 in-context source validity**, and **0.9389 composite RAG score**.
- Improved insufficient-context handling, reaching **1.00 insufficient-empty rate** on validation generation tests, enabling the model to abstain when retrieved evidence is not enough.
- Prototyped an end-to-end RAG inference system with FAISS retrieval, embedding-based chunk search, context deduplication, prompt budgeting, Qwen chat formatting, stop-token handling, and source-id parsing.
- **Tech Stack:** Python, PyTorch, Hugging Face Transformers, Datasets, PEFT, TRL, QLoRA, Qwen2.5, FAISS, Sentence Transformers